Lab Notebook, Summer 2026

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1. **Tuesday, May 19th**: Today, I spent most of my time documenting the Richards-Wolf integration code, and also building a pipeline to analyze the verification data when we take it tomorrow. There are a number of transformations required to properly compare the real and simulated data: translation in space, Poisson noise, and a potential linear (or nonlinear) scaling difference. The pipeline attempts to match these as best it can, producing a set of alignment parameters (with error bars). Then, using these parameters, we can generate many simulated images, and use them to construct a null distribution, from which we can assign a p-value to the real data. Before finishing work for the day, I also prepared a MATLAB script to run the experiment tomorrow, using the same aberrations that Diego used last summer. Finally, in anticipation of my machine learning work later in the summer, I spent an hour reading (and taking notes on) Joe Chang's *Stochastic Processes*, which I hope to complete before the end of June.

2. **Wednesday, May 20th**: From 9 AM to 1 PM, Shengqi and I worked on the 3P microscope trying to take data to check the Richards-Wolf integration code. Earlier, I had written code that would sequentially apply each of the six aberrations that Diego used last summer, and measure the PSF, a process that could be iterated as the z-level of the sample was adjusted. However, we ran into several serious issues. First, we noticed that the applied Zernike modes were not centered within the beam pupil, which we were able to quickly fix. Additionally, we had to figure out how to add a grating to the background of the applied SLM signal. Finally, we noticed that the aberrations were different in magnitude from the Python simulations. This was fixed by removing the "normalization" procedure, which was used previously to control the RMS of the aberration. In the afternoon, I finished making alterations to the code and testing it. Then, I proceeded to finish the Monte-Carlo pipeline I began yesterday, succesfully running it to completion on simulated data. However, it was noted that the resulting p-statistic is extremely sensitive to differences in the experimental data and simulation, meaning that alterations may be necessary to apply it to real data. Finally, I spent an hour taking notes on *Stochastic Processes*.

3. **Thursday, May 21st**: In the morning, I spent time making further alteration to the MATLAB code that tests the Richards-Wolf integration. While I was at it, I applied the same corrections to analogous issues with the Booth MATLAB code. These issues were non-fatal, but may have adjusted the strengths of the aberrations uniformly by an unknown amount. Then, I looked at the actual Booth data from April, and produced some plots displaying the images before vs. after correction, as well as the image metric and residual RMS over time. In the afternoon, I continued to build out the infrastructure of my utility codes, building data structures to

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organize aberrations, PSFs, and images in a more intuitive way. Hopefully, this will speed up the work going forwards. I then worked through all my Python code (including the Booth simulations from winter break) and refactored them, simplifying and abstracted a lot of messy code in the process. Over the course of several months, many changes had been made to our Booth implementation that were reflected in MATLAB but not in Python, so I also worked to bring the two scripts into alignment. Finally, I spent twenty minutes taking notes on *Stochastic Processes*.

4. **Friday, May 22nd**: From 9 AM to 1 PM, Shengqi and I again worked on the 3P microscope, trying to verify the correctness of the Richards-Wolf code. We discovered that in order to see the aberrated images, we had to decrease the strength of the aberrations and increase the laser power, often all the way up to 1.2 mW when the z-level was far away from focus. Our new code worked sucesfully! We were able to take images of a 1 micron bead under three different complex aberrations, at many different z-levels spaced by 0.2 microns, at a zoom of 70.0x. Examining the images in the afternoon, we found tight qualitative agreement between experiment and simulation. It seems like the only difference is a matter of z-calibration, and perhaps a slight x-y rotation. I tried running my quantitative comparison pipeline, but this yielded exceptionally low p-values, indicating that the code is too sensitive to minor underlying differences in the real and simulated images. I then spent two hours making the figures look nice, and organizing the results (along with the results of the Booth algorithm validation) into a Google Slides presentation. Finally, I read *Stochastic Processes* for an hour.